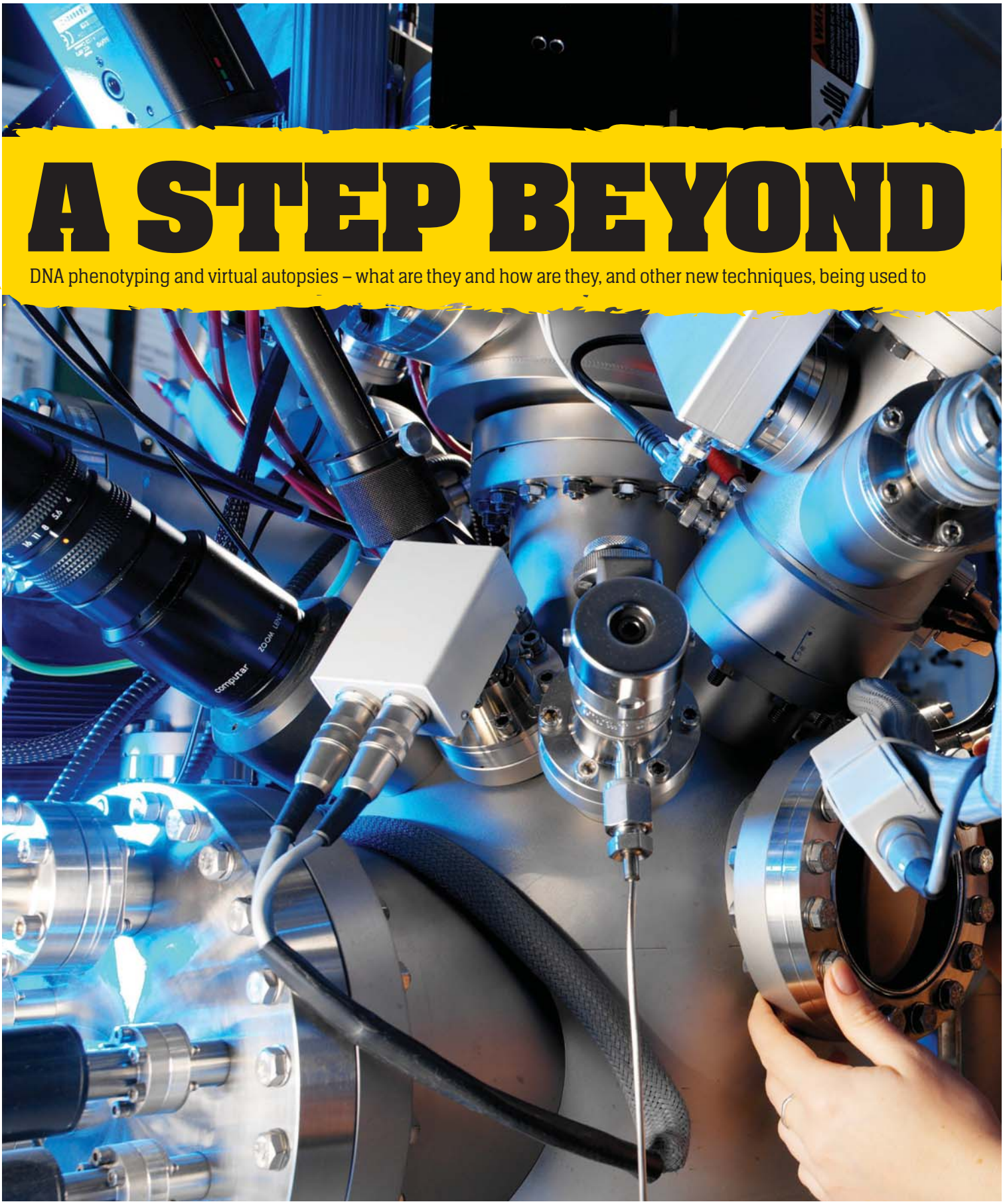


A STEP BEYOND

DNA phenotyping and virtual autopsies – what are they and how are they, and other new techniques, being used to

SCIENCE PHOTO LIBRARY

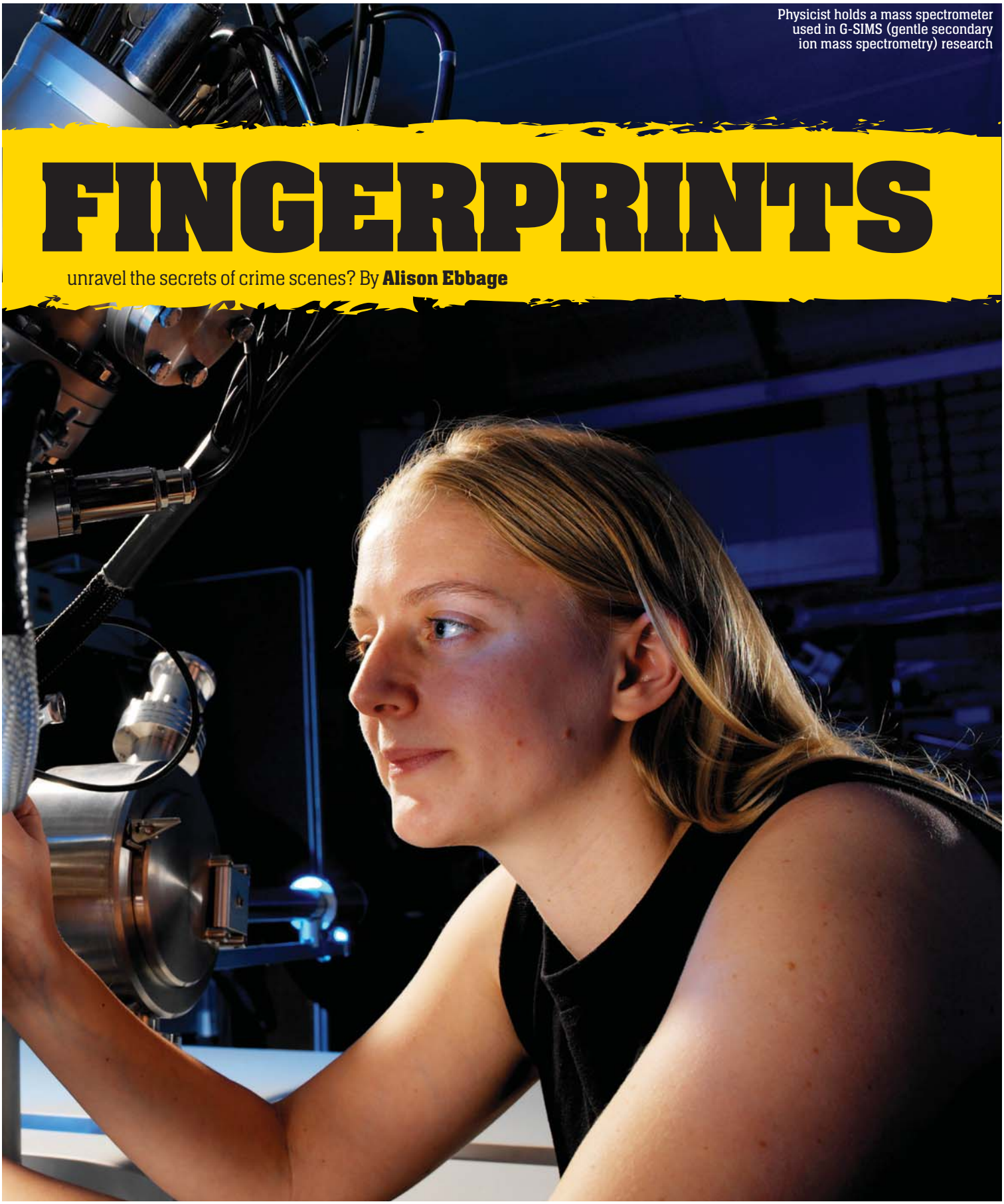
FORENSIC SCIENCE IS one of the most fascinating elements of the criminal justice system. It helps to prove cases 'beyond reasonable doubt' through the analysis of physical evidence. In everyday terms, it is the application of all the sciences to matters of law.

Such an application can help investigators understand how blood spatter patterns occur (physics), learn the composition and source of evidence such as drugs and trace materials (chemistry) or determine the identity of an unknown suspect (biology).

Professor Niamh Nic Daéid, director of the Leverhulme Research Centre for Forensic Science at the University of Dundee, comments: "Forensic science adds clear scientific rigour to the evidence process. This is important in the investigative phase; it helps to determine avenues for investigation, for example to identify or exclude a suspect. It can link cases together where there are no obvious links – such as the same footprint or the same gun used.

"It is also used within the courtroom in the evaluative phase. This is where observations are given to us by the crime scene investigators and we come up with a set of data having performed various tests on items we are given. We then provide an opinion on

Authorized licensed use limited to: COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY. Downloaded on January 25,2026 at 02:05:57 UTC from IEEE Xplore. Restrictions apply.



Physicist holds a mass spectrometer used in G-SIMS (gentle secondary ion mass spectrometry) research

FINGERPRINTS

unravel the secrets of crime scenes? By **Alison Ebbage**

the provenance of an item, or how a blood spatter pattern happened, for example. This is circumstantial evidence, and our job is to provide a scientific basis for our opinion. It needs to be robust and defendable."

This scientific and evidence-based approach adds a level of robustness to the process. Technological advances have helped to progress the degree to which experts forward a likelihood around a hypothesis.

Michael Thali, professor and chair of the Institute of Forensic Medicine at the University of Zurich and co-founder of the Virtopsy Project, says: "Forensics are the science behind solving crimes using an evidence-based approach. With forensic technology a hypothesis can be backed up which correlates to the evidence. New techniques are always down the line."

The idea is to be able to use technology to look at things in a way that is cheap, accurate and quick. With DNA, for example, in the past you would need a single profile, but now technology is more sensitive and allows for mixtures of DNA.

Analysing mixtures

Liquid chromatography mass spectrometry (LCMS) is one of the many technological advances that has been adopted in the >

Authorized licensed use limited to: COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY. Downloaded on January 25,2026 at 02:05:57 UTC from IEEE Xplore. Restrictions apply.

< forensic science world. Used in toxicology and elsewhere, it separates mixtures with multiple components. The mass spectrometry provides structural identity of individual components with high molecular specificity and detection sensitivity.

Nic Daéid says: "A good example of this in use is fire investigations where we can get samples and work out whether an ignitable liquid has been used and where to a good degree of sensitivity to allow for positive identification. In some cases, we can also point to the likely brand as they are all slightly different. The same principle applies to explosives as their residues can be looked at to identify peroxide residues."

Forensic DNA phenotyping

Another development in forensic science is an upcoming technology known as forensic DNA phenotyping (FDP). These tests provide genetic data, which can be used to infer an individual's phenotype characteristics. The data can include biological age, bio-geographical ancestry and characteristics such as hair, skin and eye colour. As of 2018, FDP in the EU is only officially regulated in Slovakia and the Netherlands.

As with LCMS, the idea is to show that a substance or element was present at a crime scene and be able to match that to something specific, be that a substance or a person. The concept can be extended using alternative light sources. The ultraviolet (UV) and visible light spectrum are used to identify substances that are not visible to the naked eye, notably trace elements.

Earlier this year, B Fakiha from the Department of Medical Health Services at Umm Al-Qura University in Saudi Arabia issued a report that supported the application of light outside the visible spectrum. "Using alternative light sources to classify biological samples is preferred because it is simple, non-destructive, and presumptive and can be used to detect a wide array of biological samples," he stated. "It is possible to identify biological samples using forensic light because of the natural characteristics that make them distinguishable from their surroundings."

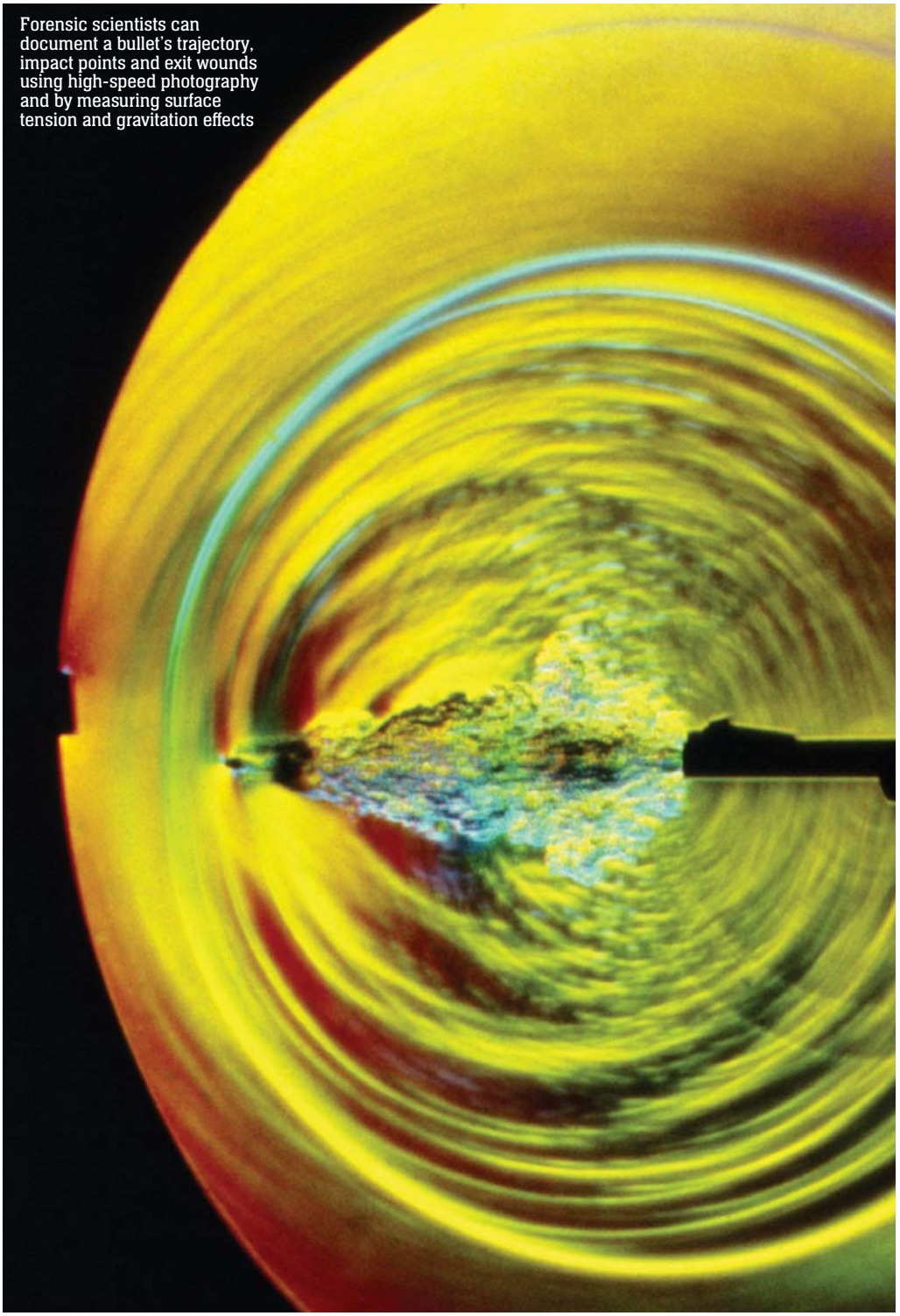
He cited blood as an example; its light absorption properties are much better. "Untreated dry blood lacks the fluorescence effect but has a high absorption across a wide range of light wavelengths, being capable of absorbing light of wavelengths between 300 and 900nm. This range encompasses the entire light wavelength including visible, ultraviolet, and infrared light. Urine, saliva and semen can (also) be identified because of their fluorescence effect."

Virtual autopsy

These technologies work to prove that an element was at a crime scene and link it to an individual or a substance. Technology that helps to show how something happened, to sequence a crime scene, or to show how injuries occurred, is also coming to the fore. The virtual autopsy is one example of this.

Thali says: "This technology scans a dead body and the crime scene in 3D to provide a 3D reconstruction – the results of which are then stored. Doing a physical autopsy destroys anything that might have been 3D so the idea is to do the scan first. We can use that data along with other elements like DNA or blood spatters to reconstruct what we think might have happened."

Nic Daéid adds: "The virtual autopsy is used to provide very high-resolution CT and/or MRI scanning to provide a 3D representation of a body. It also allows for event sequencing as to how an injury came about. The data can be stored, which means that significant inroads can be made at looking at injuries retrospectively – unlike a physical body, the results of a 3D autopsy are always available to provide new insight."



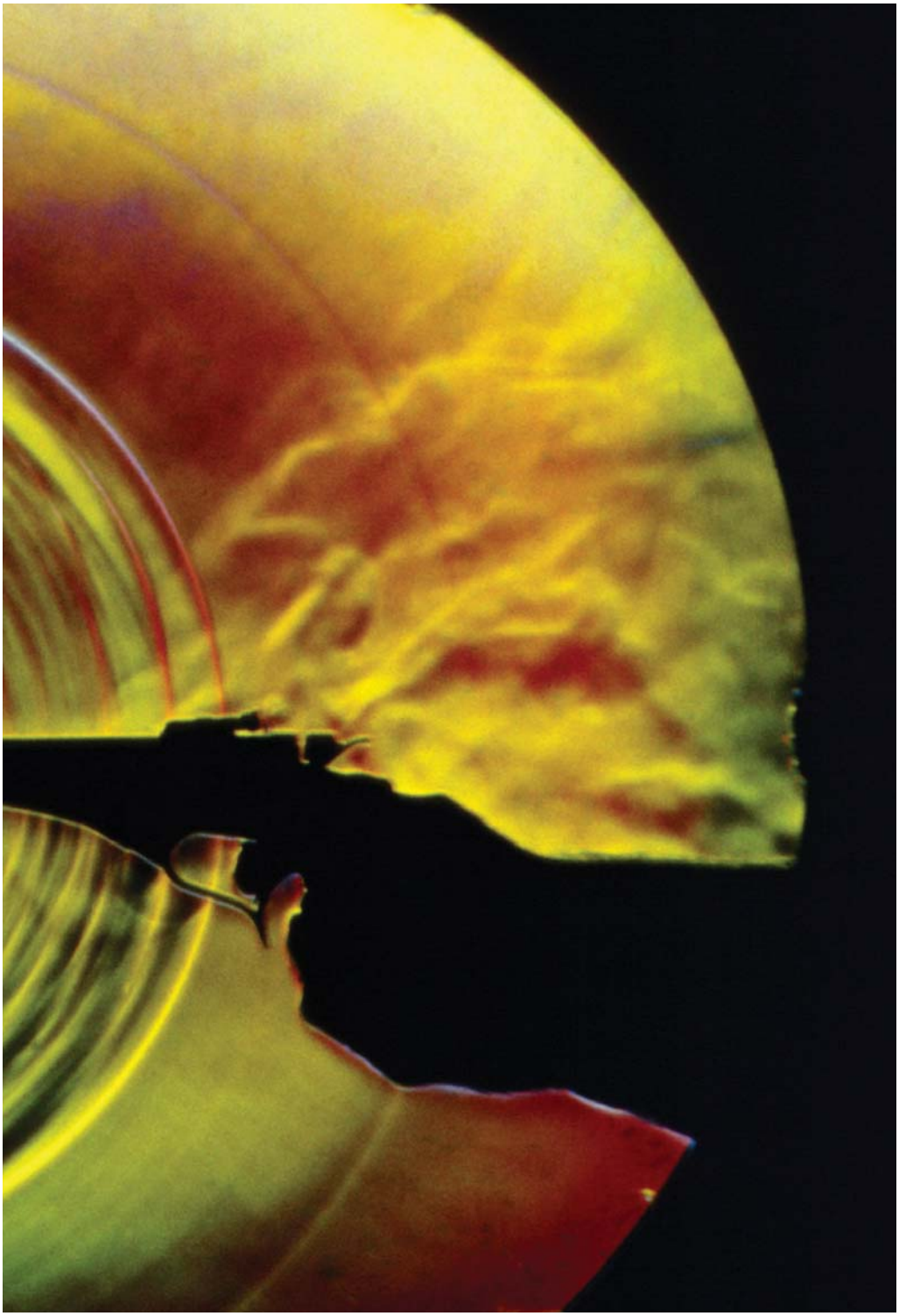
Forensic scientists can document a bullet's trajectory, impact points and exit wounds using high-speed photography and by measuring surface tension and gravitation effects

> 'Forensic science adds clear scientific rigour to the evidence process. This is important in the investigative phase; it helps to determine avenues for investigation, for example, to identify or exclude a suspect.'
> **Professor Niamh Nic Daéid** Leverhulme Research Centre for Forensic Science

High-speed photography

High-speed photography works along the same lines of building a picture of events and lends itself to ballistics forensics. This a

Authorized licensed use limited to: COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY. Downloaded on January 25,2026 at 02:05:57 UTC from IEEE Xplore. Restrictions apply.



specialised forensic science that deals with the motion, behaviour, dynamics, angular movement and effects of projectiles such as bullets, rockets, missiles, bombs etc.

For example, the examination of a bullet found at a crime scene can reveal what type of gun was used and whether it is associated with past crimes. Ballistic details are documented in a large database that is accessible by law enforcement agencies across the globe.

Using high-speed photography and measuring surface tension and gravitation effects, forensic scientists can see and document a bullet's trajectory, impact points and exit wounds. It can also be applied for military purposes to determine the accuracy of missiles and rockets and even to record the core of a nuclear explosion. However, this technique is highly reliant on how soon the film is exposed to light and so relies heavily on short but sharp flash durations.

Another application of high-speed cameras is to recreate the way that glass may have shattered at a crime scene. In turn, this works well with laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS).

This technique allows the identification of the smallest glass shards by analysing the chemical elements in their structure. It can help to prove that a glass shard found on clothing, for example, matches that found at a crime scene.

Even the smallest fragments of broken glass can reveal a lot about what happened at the scene of the crime. The LA-ICP-MS is used to analyse the glass and determine factors such as the force of impact, the direction of the bullet that broke the glass and even the type of weapon used. It does this by using highly sensitive isotopic recognition to break down glass samples of any size to their atomic structure. With this knowledge, investigators are able to match minuscule pieces of glass that have been found on clothing or other objects which may be related to the crime in some way.

Nic Daéid says: "With glass the aim is to reconstruct the glass but this can be very time-consuming, especially as most windows have a very similar type of glass, so it is hard to say which window a glass shard has come from."

Using technology to be able to say where the glass came from and how it ended up where it did can add significantly to the process of determining what happened and when at a crime scene.

Using the information

All these technologies are very useful in terms of identifying substances at a scene, linking them to something or someone else and aiming to back up a hypothesis over a sequence of events with scientific fact about, say, how a bullet does or does not travel, how blood spatters, or how glass shatters and transfers and for how long.

Nic Daéid explains: "These technologies can give us concrete information, but we then need to back that up with what we can draw from this. What does the glass shard or the blood spatter on a piece of clothing actually mean? Can we use what we know about how a material transfers onto clothing and how long it stays there to draw a conclusion about how it got there and when?"

Thali says that the biggest challenge and opportunity for these techniques lies in being able to take data from a crime scene and then use it for future reference, as well as for the crime scene it came from. "This is all about digitisation and AI being able to take large volumes of data and make informed inferences from it for future use.

"New techniques provide us with masses of data, and we need to be able to store and analyse that and present a 'beyond reasonable doubt' case. The overwhelming theme is being able to harness digitisation to process the vast amounts of data we can get from those techniques so as to be able to store, analyse and interrogate that data and share it when necessary or relevant."

Building data sets that can be seen by all so as to provide a bank of valuable data is a key part of this and already in use within the toxicology field. Nic Daéid says: "With liquid chromatography mass spectrometry (LCMS), it is all about building a robust record of as many data sets as possible that we can then refer back to and compare and contrast with debris samples."

The crux of the issue seems to be harnessing new technologies and being able to apply them to forensic science in a way that is acceptable to a court of law. Being able to do this going forward, it seems, means keeping pace with the broader world in terms of digitisation and how data is collected, stored and analysed. To maintain a world-leading legal system that is supported by forensic science, further investment in keeping things up to date and leading with new technology will now, perhaps, become a focus. *

Authorized licensed use limited to: COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY. Downloaded on January 25,2026 at 02:05:57 UTC from IEEE Xplore. Restrictions apply.